MX-200 Preventive Maintenance Manual

Inspection, replacement, restart, and maintenance-record procedures

Classification: Internal - All Employees

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Document owner	Maintenance Engineering
Intended audience	Trained maintenance technicians, supervisors, and reliability engineers
Fixture status	Synthetic evaluation document

Document purpose

This controlled document provides realistic, structured content for the UC03 role-aware document retrieval and question-answering evaluation. It is not an operational company instruction.

Contents

1. Safety

Apply lockout-tagout before opening the MX-200 filter housing. Only trained maintenance staff may perform the work.

The MX-200 contains pressurised process media and rotating auxiliary equipment. Maintenance must be performed under the site permit system and the current equipment isolation procedure. Never rely only on the operator display to confirm a zero-energy state.

Hazard	Control
Stored pressure	Isolate upstream and downstream valves, vent to the approved collection point, and verify zero pressure.
Unexpected start	Apply electrical and process lockout-tagout and test the isolation before work.
Hot surface	Allow the housing to cool and use task-appropriate gloves.
Process residue	Use the specified eye, hand, and protective-clothing controls for the handled material.
Heavy cover	Use the designated lifting point and keep hands clear of the sealing face.

1.1 Stop-work conditions

Stop the task and notify the maintenance supervisor if isolation cannot be verified, the housing is damaged, the process medium is unknown, or the approved replacement element is unavailable. Temporary bypass operation is not authorised by this manual.

2. Equipment Overview

The MX-200 filtration assembly protects the main process loop from particulate contamination. Its serviceable elements are the main process filter, housing seal, differential-pressure gauge, vent connection, drain connection, and cover fasteners.

Item	Function	Routine observation
Main process filter	Captures process particles	Condition, deposits, damage, and service hours
Differential-pressure gauge	Indicates restriction across filter	Stable reading and readable scale
Housing seal	Maintains pressure boundary	Cuts, flattening, swelling, or contamination
Vent and drain	Supports safe depressurisation	Clear route, closed after service

3. Preventive Maintenance Programme

Operating hours are read from the MX-200 service counter and recorded in the maintenance log. Calendar checks support safe operation but do not replace the operating-hour intervals below. A supervisor may shorten an interval when process conditions are unusually dusty or unstable.

Maintenance item	Trigger	Required record
External condition check	At the start of each planned service visit	Leaks, corrosion, gauge condition, and housekeeping
Main process filter inspection	Every 250 operating hours	Hours, differential pressure, visible condition, and technician
Main process filter replacement	At 1,000 operating hours or above the pressure limit, whichever occurs first	Element lot, removed-element condition, seal status, and restart values
Housing-seal assessment	Whenever the housing is opened	Reuse or replacement decision and observed condition

4. Filter Inspection

Inspect the main process filter every 250 operating hours. Record the differential pressure and visible condition in the maintenance log.

4.1 Preparation

Review the previous log entry, confirm the current operating-hour reading, identify any process-quality complaints, and obtain the approved inspection tools. If the housing will be opened, complete the full isolation and depressurisation sequence in Section 1.

4.2 Inspection sequence

- Confirm equipment identity and record the service-counter value.
- Read and record the differential pressure under stable operating conditions before shutdown when safe to do so.
- Check the housing, gauge, vent, drain, and connections for leakage or damage.
- When an internal inspection is required, examine the filter surface, end caps, seating area, and seal for deposits, deformation, tears, or bypass marks.
- Record whether continued service, planned replacement, or immediate replacement is required.

4.3 Inspection decision

A filter may remain in service only when the element is intact, the pressure reading is within the permitted range, and no bypass evidence is present. Damage, collapse, missing seal contact, or an unreadable pressure gauge requires supervisor review before restart.

5. Filter Replacement

Replace the main process filter after 1,000 operating hours or earlier when differential pressure exceeds 1.8 bar, whichever occurs first.

The operating-hour and pressure triggers are independent. Reaching either trigger requires replacement even when the other value remains within its normal range. A visibly damaged or collapsed element must also be replaced immediately.

5.1 Required materials

Item	Requirement
Replacement filter	Approved MX-200 element with verified part number and intact packaging
Housing seal	Approved seal material compatible with the process medium
Cleaning materials	Lint-free wipes and site-approved cleaning agent
Tools	Calibrated torque tool and designated cover-lifting equipment
Records	Current work order and maintenance-log entry

5.2 Replacement procedure

- Stop the MX-200, isolate all energy sources, depressurise, drain, and verify the safe state.
- Open the housing using the specified lifting point. Protect the sealing faces from tools and debris.
- Remove the used element without dislodging deposits into the clean side of the housing.
- Inspect and clean the element seat, housing interior, cover, and seal groove.
- Install the approved element in the correct orientation and verify full seating.
- Inspect the housing seal and replace it when damaged, deformed, chemically affected, or outside its approved reuse condition.
- Close the housing and tighten fasteners evenly using the controlled sequence on the equipment label.
- Record the replacement element, lot number, service hours, cause of replacement, and technician name.

5.3 Disposal

Handle the removed filter according to the process-medium classification and site waste procedure. Do not place contaminated elements in general waste unless the approved waste classification explicitly permits it.

6. Restart Check and Functional Verification

After replacement, confirm that the housing is sealed and differential pressure is below 0.5 bar before returning the unit to service. Reintroduce pressure gradually and observe the housing, vent, drain, gauge, and process connections throughout the restart.

Check	Acceptance condition
Housing closure	Cover is seated, fasteners are secured, and the seal is not displaced
Vent and drain	Closed and free from leakage
Differential pressure	Below 0.5 bar after stabilisation
External leakage	No visible seepage at the cover or connections
Process response	Stable flow and no abnormal vibration or noise
Record completion	Restart readings and observations entered in the maintenance log

7. Troubleshooting

Observation	Possible cause	Action
Pressure rises quickly after replacement	Incorrect element, process upset, or severe contamination	Stop safely, verify the element specification, inspect the element, and notify process supervision.
Leak at housing cover	Seal damage, contamination, misalignment, or uneven closure	Stop, isolate, depressurise, inspect sealing surfaces, and reinstall with an approved seal.
Gauge remains at zero	Blocked impulse path, damaged gauge, or no process flow	Verify process status and gauge function before relying on the reading.
Particles downstream	Element damage, poor seating, or bypass path	Remove the unit from service and inspect the element and seating surfaces.

Do not reset the service counter or close the work order until the restart checks and maintenance record are complete.

8. Records, Spares, and Escalation

8.1 Maintenance record

Each inspection and replacement record must identify the MX-200 asset, date, operating hours, differential pressure, observed condition, action taken, parts used, technician, and approving supervisor when escalation was required.

Field	Example entry format
Asset	MX-200 / site asset identifier
Operating hours	Numeric service-counter reading
Differential pressure	Measured value in bar and operating state
Condition	Clean, loaded, damaged, collapsed, bypass marks, or other observation
Action	Continue service, replace, repair, or escalate
Parts	Part number and lot or batch identifier

8.2 Spare-part control

Store replacement elements in clean, dry packaging away from chemical vapours and mechanical damage. Before installation, verify the part number, shelf condition, packaging integrity, and compatibility with the process assignment.

8.3 Escalation criteria

- Repeated early loading or unexplained pressure increase after replacement.
- Evidence of filter collapse, bypass, housing damage, or downstream contamination.
- Disagreement between service-counter history and maintenance records.
- Unavailable approved parts or inability to verify safe isolation.
- Any condition not covered by the approved maintenance procedure.

Revision history

Version	Date	Summary
2.2	10 Mar 2025	Added inspection decision guidance.
2.3	20 Feb 2026	Expanded restart and record controls.
2.4	1 Jul 2026	Clarified controlled-document metadata and escalation examples.